

Real-World Order of Operations

Question Bank 4.07

Name _____

Date _____

Period _____

Show your work. Circle your final answer.

For questions 1 through 4, substitute and evaluate.

1. $4x^2$ when $x = -3$

2. $\frac{1}{2}mv^2$ when $m = 50$ and $v = 6$

3. $\frac{a+b^2}{c}$ when $a = 6$, $b = 5$, $c = 2$

4. $2(x-y)^2$ when $x = 3$ and $y = 7$

For questions 5 and 6, write and evaluate an expression. Give the unit.

5. A square room of side 11 feet is floored at 3.20 dollars a square foot, with a flat 75 dollars for delivery. Find the total cost.

6. A rectangular deck 14 feet by 9 feet is stained at 1.15 dollars a square foot, with a flat 48 dollars for equipment hire. Find the total cost.

Two formulas.

7. Body mass index is $B = \frac{703w}{h^2}$, where w is a weight in pounds and h is a height in inches. Find B to the nearest tenth.

a. $w = 150$, $h = 65$

b. $w = 180$, $h = 70$

c. $w = 200$, $h = 72$

8. Kinetic energy is $K = \frac{1}{2}mv^2$, for mass m in kilograms and speed v in meters per second. Find K in joules.

a. $m = 1,200$, $v = 15$

b. $m = 850$, $v = 12$

c. $m = 70$, $v = 8$

Working backwards.

9. Find the missing value in each situation.

a. $\frac{1}{2}mv^2 = 900$ and $v = 6$. Find m .

b. $x^2 + 5 = 41$, where x is positive. Find x .

- c. A square lawn of side n feet is turfed at 2.50 dollars a square foot with a flat 40 dollars delivery, and the total is 2,290 dollars. Find n .

Reading the situation.

10. Two problems use the numbers 20, 3 and 5. **A:** a gym charges 20 dollars to join plus 3 dollars for each of 5 classes. **B:** a gym charges 20 dollars for each of 3 months, plus 5 dollars to join. Find both totals, and explain why the words 'plus' and 'each' do not decide the operation.
11. A square yard of side 15 feet is turfed at 4.80 dollars a square foot, with a flat 120 dollars for delivery. Which expression gives the total cost?
- ☐ A. $15(4.80) + 120$
 - ☐ B. $15^2(4.80) + 120$
 - ☐ C. $15^2(4.80 + 120)$
 - ☐ D. $15^2 + 4.80 + 120$
12. Select ALL of the expressions whose value is 45 when $x = 3$.
- ☐ A. $(5x)^2$
 - ☐ B. $\frac{135}{x}$
 - ☐ C. $5x^2$
 - ☐ D. $3x^2 + 18$
 - ☐ E. $5x \cdot 2$
 - ☐ F. $x^2 + 36$